Now $\dfrac{\partial g}{\partial r} = \dfrac{\partial g}{\partial x}\dfrac{\partial x}{\partial r} + \dfrac{\partial g}{\partial y}\dfrac{\partial y}{\partial r} = 2x(2) + 2(2r) = 12r - 4s$.

also, $\dfrac{\partial g}{\partial s} = \dfrac{\partial g}{\partial x}\dfrac{\partial x}{\partial s} + \dfrac{\partial g}{\partial y}\dfrac{\partial y}{\partial s} = 2x(-1) + (2)2 = 2s - 4r + 4$.

EXERCISE 8.6

1. If $u(x, y) = x^2y + 3xy^4$, $x = e^t$ and $y = \sin t$, find $\dfrac{du}{dt}$ and evaluate it at $t = 0$.
2. If $u(x, y, z) = xy^2z^3$, $x = \sin t$, $y = \cos t$, $z = 1 + e^{2t}$, find $\dfrac{du}{dt}$.
3. If $w(x, y, z) = x^2 + y^2 + z^2$, $x = e^t$, $y = e^t \sin t$ and $z = e^t \cos t$, find $\dfrac{dw}{dt}$.
4. Let $U(x, y, z) = xyz$, $x = e^{-t}$, $y = e^{-t}\cos t$, $z = \sin t$, $t \in \mathbb{R}$. Find $\dfrac{dU}{dt}$.
5. If $w(x, y) = 6x^3 - 3xy + 2y^2$, $x = e^s$, $y = \cos s$, $s \in \mathbb{R}$, find $\dfrac{dw}{ds}$, and evaluate at $s = 0$,
6. If $z(x, y) = x\tan^{-1}(xy)$, $x = t^2$, $y = se^t$, $s, t \in \mathbb{R}$. Find $\dfrac{\partial z}{\partial s}$ and $\dfrac{\partial z}{\partial t}$ at $s = t = 1$.
7. Let $U(x, y) = e^x \sin y$, where $x = st^2$, $y = s^2t$, $s, t \in \mathbb{R}$. Find $\dfrac{\partial U}{\partial s}, \dfrac{\partial U}{\partial t}$ and evaluate them at $s = t = 1$.
8. Let $z(x, y) = x^3 - 3x^2y^3$, where $x = se^t, y = se^{-t}$, $s, t \in \mathbb{R}$. Find $\dfrac{\partial z}{\partial s}$ and $\dfrac{\partial z}{\partial t}$.
9. $W(x, y, z) = xy + yz + zx$, $x = u - v$, $y = uv$, $z = u + v$, $u, v \in \mathbb{R}$. Find $\dfrac{\partial W}{\partial u}, \dfrac{\partial W}{\partial v}$, and evaluate them at $\left(\dfrac{1}{2}, 1\right)$.

8.6.2 Homogeneous Functions and Euler's Theorem

Definition 8.12

(a) Let $A = \{(x, y) \mid a < x < b,\ c < y < d\} \subset \mathbb{R}^2$, $F : A \to \mathbb{R}$, we say that F is a homogeneous function on A, if there exists a constant p such that $F(\lambda x, \lambda y) = \lambda^p F(x, y)$ for all $\lambda \in \mathbb{R}$ and sutitably restricted $\lambda, x, y,$ such that $(\lambda x, \lambda y) \in A$. This constant p is called degree of F.

(b) Let $B = \{(x, y, z) \mid a < x < b,\ c < y < d,\ u < z < v\} \subset \mathbb{R}^3$, $G : B \to \mathbb{R}$, we say that G is a homogeneous function on B, if there exists a constant p such that $G(\lambda x, \lambda y, \lambda z) = \lambda^p G(x, y, z)$ for all $\lambda \in \mathbb{R}$ and sutitably restricted $\lambda, x, y, z,$ such that $(\lambda x, \lambda y, \lambda z) \in B$. This constant p is called degree of G.

Note: Division by any variable may occur, to avoid division by zero, we say that λ, x, y, z are sutitably restricted real numbers.

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